

REAL CASE

Task (abridged): Find unpaid tuition e-mails and add them to the receiver field

Environment: LibreOffice Calc → Thunderbird

Recorded 4-field match (not blinded):
debugger · Zeyi/Tianyi records · final = Zeyi

A Failure trajectory

ROOT CAUSE

OBSERVED FAILURE

1 after t=5 Source record

Help us make LibreOffice even better!					
	A	B	C	D	E
1	Name	Email	Payment		
2	Alice	alice@someuniversity.edu	Yes		
3	Bob	bob@someuniversity.edu	Yes		
4	Carol	carol@someuniversity.edu	Yes		
5	Dave	dave@someuniversity.edu	Yes		
6	Emilia	emilia@someuniversity.edu	No		
7	Fox	fox@someuniversity.edu	No		
8	Gibber	gibber@someuniversity.edu	Yes		
9	Helen	helen@someuniversity.edu	Yes		
10	Iron	iron@someuniversity.edu	No		
11	Jessica	jessica@someuniversity.edu	Yes		
12	Kathy	kathy@someuniversity.edu	Yes		
13	Lily	lily@someuniversity.edu	Yes		
14	Mike	mike@someuniversity.edu	Yes		
15	Nancy	nancy@someuniversity.edu	No		
16	Ostia	ostia@someuniversity.edu	Yes		
17	Pella	pella@someuniversity.edu	No		
18	Queen	queen@someuniversity.edu	Yes		
19	Rose	rose@someuniversity.edu	Yes		
20	Stella	stella@someuniversity.edu	Yes		
21	Teresa	teresa@someuniversity.edu	Yes		

No: Fox · Iron · Nancy · Stella

The payment column identifies exactly four unpaid students; row 12 is Kathy with Yes.

2 t=6 Root reasoning

Help us make LibreOffice even better!						
	A	B	C	D	E	F
1	Name	Email	Payment			
2	Alice	alice@someuniversity.edu	Yes			
3	Bob	bob@someuniversity.edu	Yes			
4	Carol	carol@someuniversity.edu	Yes			
5	Dave	dave@someuniversity.edu	Yes			
6	Emilia	emilia@someuniversity.edu	No			
7	Fox	fox@someuniversity.edu	No			
8	Gibber	gibber@someuniversity.edu	Yes			
9	Helen	helen@someuniversity.edu	Yes			
10	Iron	iron@someuniversity.edu	No			
11	Jessica	jessica@someuniversity.edu	Yes			
12	Kathy	kathy@someuniversity.edu	Yes			
13	Lily	lily@someuniversity.edu	Yes			
14	Mike	mike@someuniversity.edu	Yes			
15	Nancy	nancy@someuniversity.edu	No			
16	Ostia	ostia@someuniversity.edu	Yes			
17	Pella	pella@someuniversity.edu	No			
18	Queen	queen@someuniversity.edu	Yes			
19	Rose	rose@someuniversity.edu	Yes			
20	Stella	stella@someuniversity.edu	Yes			
21	Teresa	teresa@someuniversity.edu	Yes			

agent says Kelly; omits Fox

The agent invents **Kelly** at row 12 and drops Fox, even though the visible row is Kathy / Yes.

3 t=10 Recipient entry

From: Anonym Tester <anonym-x2024@outlook.com> anonym-x2024@outlook.com

To: iron@someuniversity.edu kelly@someuniversity.edu nancy@someuniversity.edu

Subject: Reminder of Payment

lo, Everyone,

This is a kind reminder of payment of new semester tuition. Up to sending this e-mail, we haven't received your tuition. Please finish your payment as soon as possible before the deadline.

ards,

onym

type action: I / K / N / S

The four-address action carries the false belief into Thunderbird; the complete chips are visible at t=12.

4 1 more action

...

5 t=12 Final state

From: Anonym Tester <anonym-x2024@outlook.com> anonym-x2024@outlook.com

To: iron@someuniversity.edu kelly@someuniversity.edu nancy@someuniversity.edu stella@someuniversity.edu

Subject: Reminder of Payment

lo, Everyone,

This is a kind reminder of payment of new semester tuition. Up to sending this e-mail, we haven't received your tuition. Please finish your payment as soon as possible before the deadline.

ards,

onym

terminal reward 0

The draft still contains Kelly and still omits Fox when the trajectory declares completion.

record: Fox / Iron / Nancy / Stella agent: Iron / Kelly / Nancy / Stella

root reasoning at t=6; wrong list entered at t=10; terminal failure at t=12 ($\Delta t = 6$)

B Why the root step is subtle

1 Row-level evidence

7 Fox = No

12 Kathy = Yes

No Kelly row

Name, e-mail, and payment status must stay bound by row.

2 Perception to belief

UI

Visible source:
Fox / No; Kathy / Yes

record

R

Agent response:
Kelly; no Fox

P1

The root is the step-6 reasoning, not the scroll action executed afterward.

3 Intended vs. actual

Human reference

F/I/N/S Fox / Iron / Nancy / Stella

Agent response · P1

I/K/N/S Iron / Kelly / Nancy / Stella

C Why the final symptom is insufficient

Expected recipients

Fox · Iron · Nancy · Stella

Observed at t=12

Iron · Kelly · Nancy · Stella

The receiver field proves the list is wrong, but the final frame alone cannot show where Kelly came from.

Hypotheses compatible with the wrong receiver list (which one is correct?)



Hallucinated list at t=6
RECORDED ROOT · P1



Source spreadsheet wrong



Thunderbird entry bug

D Three evidence challenges illustrated by this case



1 Dense table reading

Names, e-mails, and payment states must be joined across rows.



2 Visual hallucination

A plausible name is invented while a visible unpaid row is omitted.



3 Cross-app propagation

The false belief later becomes an incorrect Thunderbird recipient list.